EAS 487

HOMEWORK 2

SOLUTIONS

1. $H_\phi = I\,dl \sin\theta \frac{1}{4\pi}\left[\frac{jk_0}{r} + \frac{1}{r^2}\right] e^{j(\omega t - k_0 r)}$

$E_r = -I\,dl\cos\theta \frac{j\eta_0}{2\pi k_0}\left[\frac{jk_0}{r^2} + \frac{1}{r^3}\right] e^{j(\omega t - k_0 r)}$

$E_\theta = I\,dl \sin\theta \frac{j\eta_0}{4\pi k_0}\left[\frac{k_0^2}{r} - \frac{jk_0}{r^2} - \frac{1}{r^3}\right] e^{j(\omega t - k_0 r)}$

$\langle P_r \rangle = \frac{1}{2}\,\mathrm{Re}\,\underline{E}\times\underline{H}^* \cdot \hat{r}$ GIVES AVERAGE POWER DENSITY PASSING THROUGH A SPHERICAL SURFACE. ONLY $E_\theta H_\phi^*$ PRODUCT CONTRIBUTES

$$= \frac{1}{2}\mathrm{Re}\,|I|^2 dl^2 \sin^2\theta \frac{1}{(4\pi)} \frac{j\eta_0}{4\pi k_0}\underbrace{\left(\frac{k_0^2}{r} - \frac{jk_0}{r^2} - \frac{1}{r^3}\right)\left(-\frac{jk_0}{r} + \frac{1}{r^2}\right)}_{-\frac{jk_0^3}{r^2} + \cancel{\frac{k_0^2}{r^3}} - \cancel{\frac{k_0^2}{r^3}} - \cancel{\frac{jk_0}{r^4}} + \cancel{\frac{jk_0}{r^4}} \underbrace{- \frac{1}{r^5}}_{\text{CONTRIBUTES TO IMAG. PART}}}$$

$$= \frac{1}{2}|I|^2 dl^2 \sin^2\theta \frac{\eta_0}{(4\pi)^2}\frac{k_0^2}{r^2}$$

TOTAL POWER RADIATED $= \int_{\text{SPHERE}} \langle P_r \rangle r^2 \overbrace{d\Omega}^{\sin\theta\, d\theta\, d\phi} = \int_0^{2\pi}\int_0^{\pi} \frac{1}{2}|I|^2 dl^2 \sin^3\theta \frac{\eta_0}{(4\pi)^2} k_0^2\, d\theta\, \underbrace{d\phi}_{\to 2\pi}$

HAVE TO PERFORM $\int_0^\pi \sin^3\theta\, d\theta = \int_0^\pi (1-\cos^2\theta)\sin\theta\, d\theta = \cos\theta - \frac{\cos^3\theta}{3}\Big|_\pi^0$

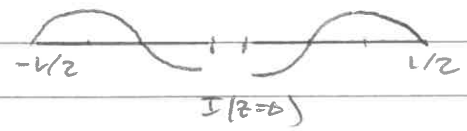
$= 4/3$

SO WE GET $\frac{1}{2}\frac{4}{3}\,2\pi\,|I|^2 dl^2 \frac{\eta_0}{(4\pi)^2} k_0^2$ $\quad k_0 = \frac{2\pi}{\lambda}$ $\quad \eta_0 = 120\pi$

$= 40\pi^2 |I|^2 (dl/\lambda)^2$ W

* NOTE: MAX DIRECTIVITY ~ 3.3 FOR $L \cong 5\lambda/4$ DIPOLE

2 $L = 3\lambda/2$ $\quad I(z) = I_0 \sin\left[k(L/2 - |z|)\right]$ $\quad k\frac{L}{2} = \frac{2\pi}{\lambda}\frac{3\lambda}{4} = \frac{3\pi}{2}$



NOTE: FEEDPOINT CURRENT NOT NECESSARILY I_0

RATHER, $I_f = I_0 \sin\frac{3\pi}{2}$ OR $-I_0$ HERE

HAVE $\langle P_r \rangle = \frac{Z_0}{2}\frac{|I_0|^2}{(2\pi r)^2}\left[\frac{\cos(3\pi/2\cos\theta - \cos 3\pi/2)}{\sin\theta}\right]^2$

$$P_+ = \int \langle P_r \rangle r^2 d\Omega = \frac{Z_0|I_0|^2}{8\pi^2}\underbrace{\int_0^{2\pi} d\phi}_{2\pi}\underbrace{\int_0^{\pi}\left[\quad\right]^2 \sin\theta\, d\theta}_{1.758}$$

$$= 1.758\,\frac{Z_0|I_0|^2}{4\pi}\ \text{W}$$

$$D = \frac{\langle P_r \rangle}{P_+/4\pi r^2} = \frac{2}{1.758}\left[\frac{\cos(3\pi/2\cos\theta)}{\sin\theta}\right]^2 = D(\theta,\phi)$$

MAX VALUE = 1 AT $\theta = \pi/2$

SO $A_{MAX} = 2.22$ $(\theta \sim 42°)$ $\quad$ NO - MAX AT ? $\theta = 0.742$

↳ 1.957

RADIATION RESISTANCE $\quad P_+ = 1.758\,\frac{Z_0|I_0|^2}{4\pi} = \frac{1}{2}I_f^2 R_a$ $\quad$ NOT BROADSIDE

↑ FEEDPOINT CURRENT

SO $R_a = 0.28\,Z_0 = 105\ \Omega$ $\quad = \frac{1}{2}I_0^2 R_a$ IN THIS CASE

EFFECTIVE AREA $\quad A_{eff}(\theta,\phi) = \frac{\lambda^2}{4\pi}D(\theta,\phi)$

HPBW FIND NUMERICALLY - CONSIDER ONE OF THE FOUR MAIN BEAMS

HALF-POWER POINTS AT 0.42, 1.00 RADIANS

SO HPBW = $(1.00 - 0.42)\frac{180}{\pi} = 33.2°$

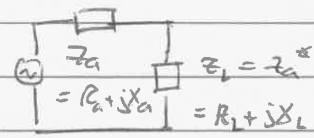
SMALLER THAN $\lambda/2$ DIPOLE

3 $\langle P_{inc} \rangle = \frac{|E|^2}{2\eta_0} \cong \frac{1}{2 \cdot 377\Omega} = 1.32 \text{ mW/m}^2$ INCIDENT POWER DENSITY

USING RECIPROCITY, $A_{eff} = \frac{\lambda^2}{4\pi} D$ $D = 1.5$ FOR SHORT DIPOLE

SO $\langle P_r \rangle = \langle P_{inc} \rangle A_{eff} = 1.32 \times 10^{-3} \frac{\lambda^2}{4\pi} D = 1.58 \times 10^{-4} \lambda^2 \text{ W}$

NOW, WITHOUT RECIPROCITY



MATCHED LOAD

$V_{oc} = \underline{E} \cdot \underline{h} = 1 \text{ mV/m} \cdot dl \cdot \frac{1}{2}$ short dipole

$P_{rx} = \frac{1}{2} I^2 R_L = \frac{1}{2} \frac{|V_{oc}|^2}{|R_a + Z_L|^2} R_L = \frac{1}{2} \frac{|V_{oc}|^2}{(2R_a)^2} R_a = \frac{1}{8} \frac{|V_{oc}|^2}{R_a}$

$R_a \cong 80\pi^2 \left(\frac{dl}{\lambda}\right)^2 \cdot \frac{1}{4}$ ← short dipole

$= \frac{1}{8} \frac{dl^2}{80\pi^2} \left(\frac{\lambda}{dl}\right)^2$

$\cong 1.58 \times 10^{-4} \lambda^2 \text{ W}$

4 EARTH ~~to~~ $r = 2\times10^{13}$ m VOYAGER

$P_t = 20$ W $\quad A_{eff} = 0.55\,\pi\left(\frac{3.7}{2}\right)^2 = 5.9\ m^2$

$G = 4\pi\frac{A_{eff}}{\lambda^2} = 8.24\times10^4$ (49 dB)

INCIDENT POWER DENSITY AT EARTH $\quad P_{inc} = P_t\frac{G}{4\pi r^2} = 3.27\times10^{-22}$ W/m^2

(SMALL!)

NOISE POWER $\quad P_N = k_B B T = 1.9\times10^{-20}$ W

FOR 10 dB SNR, REQUIRE $\quad P_r = 1.9\times10^{-19}$ W

$= P_{inc}\cdot A_{eff}$

SO $\quad P_x\ A_{eff} = 5.8\times10^{2}\ m^2$

$A_{dishes} = A_{eff}/0.55 = \pi R^2 \rightarrow R = 18.3$ m

NASA'S DEEP SPACE NETWORK INCLUDES THE GOLDSTONE DISH WHICH HAS A 35-M RADIUS, SO THIS CAN WORK (FOR NOW)